Contents

The Completion **1**
1. The action . . . 3
2. What is verified . . . 6
3. Parameter count, honestly . . . 13
4. What is NOT claimed — read this before quoting anything above . . . 13
5. The falsifiable predictions . . . 17
6. Credit . . . 20

The Completion

A relativistic field theory carrying $a_0 = \kappa c \sqrt{(G\rho_\Lambda)}$

Zimmerman, Carl P. — Briar Creek Tech Assembled 2026-08-09; **v2, v3, v4, v5 same day; v6 2026-08-10; v7 2026-08-10 (late); v8 2026-08-10 (night, stages 19–22); v9 2026-08-11; v10 2026-10-12 (errata only). v10 ERRATA — read before citing anything below. This document is no longer a viable field theory; it is kept as the record of the 2026-08-11 state.** Four later results close pillars of v9. Nothing below has been rewritten; the text stands as v9 published it. 1. **The single-field dark sector is EXCLUDED.** "The DBI khronon's dust is Ω_dm" (§1.2) pins $R = \Lambda_D/Q_0 = 2.6\nu_0$ through $\beta = 1$, 18–300× above the P(k = 0.2) ceiling; P(k = 0.2) is suppressed 44–100%. A separate pressure-supported field cannot keep Ω_dm out of galaxies for any K (`qwen_claude_field_theory/closure_2026/condensate_pincer_2026/condensate_mu_pincer_2026.py`). 2. **The static limit fails the Solar System as modified gravity, and lensing as modified inertia.** The kernel of §1.1 (*νRAR, correctly transcribed here;* ***`hunt_2026/f23_kernel_transcription_audit.py`***) *gives a Solar-System EFE quadrupole 6.2×/6.8× (QUMOND) over the Cassini/Park ceiling (*`hunt_2026/f24_aqual_quadrupole_rar_kernel.py`*); no one-argument law $\mu(g/a_0)$ passes both galaxies and Cassini (*`hunt_2026/f28_one_argument_pincer.py`*). The modified-inertia reading is excluded by lensing at ~21σ (*`real_research/reviews/mi_mg_arm_standing_2026.py`*). Any completion needs a second argument that is a length; none is written here. 3.* ***The a_0–dark-energy link of §1.3 is a TARGET relation, not a derivation.*** *The promotion $a_0^2(Q) \propto -K(Q)$ is a definitional choice. It reproduces $-K = \rho\Lambda$ only as an identification at the de Sitter minimum (w = -1, a_0 constant); an evolving $a_0(z) \propto \sqrt{\rho_DE(z)}$ is not derived from the action* (the open inverse-K(Q) problem). 4. $\kappa = \frac{1}{2}$ **is fitted** (already non-claim 1; repeated because it carries every number). **What survives:** the a_0-line phenomenology $g = \sqrt{(g_N^2 + g_N\, a_0)}$, its fit to SPARC, and the falsifiable $a_0(z)$ prediction — flat to <1% below $z \approx 5$ against the $a_0 \propto H(z)$ rival. §5's collapse speedup stands only as the a_0-line calculation it is (1.96× an upper bound); the theory that would source it is items 1–3. Never cite this document as a working relativistic completion. **v9 changes — one prediction RE-DERIVED (its mechanism was misattributed), two new predictions added, and an S8 null.** §5's "accelerated structure formation / earlier massive objects" row is **RESTORED and re-derived**. A draft of this version withdrew it, arguing that abundance is set by linear growth and that rows 19–20 make linear growth ΛCDM's. That reasoning was **wrong in substance**: collapse *timing* is nonlinear, the a_0-line governs accelerations at every scale, and the $\delta Y^1 = 0$ theorem says nothing about nonlinear collapse. The two facts coexist — **ΛCDM halo abundance AND accelerated collapse timing.** The draft compounded the error by pricing the nonlinear boost at $\delta \approx 200$

(the *end* of collapse, $\lesssim 1.5\times$) instead of at turnaround (**2.3–5.1**×), which is what sets the timing. Computed from the a_0-line alone: **collapse is 1.34–1.96× faster than Newtonian**, so objects ΛCDM forms at z = 10 appear at z ≈ 14–16 — and because $a_0(z)$ declines, the speedup declines with it (2.03× at z = 6 → 1.14× at z = 25), a shape no constant-a_0 MOND predicts. The double-counting fork (the boost acts on total enclosed mass) is named as the live uncertainty, making 1.96× an upper bound. ALSO ADDED, from the framework's *own* two equations with zero free parameters: the **early-rotator BTFR zero point** $v(z)/v(0) = [a_0(z)/a_0(0)]^{1/4}$ — flat to 0.009% below z = 5, then a signed velocity deficit reaching 7.8% at z = 20, testable once it clears the 3.5% floor at z ≈ 17 — and the framework's **own** MOND threshold $\Sigma = a_0/(\pi G) = 214$ M/pc^2 (twice the value usually quoted, which belongs to a different interpolation), which *shrinks* toward high z. Also recorded: the χ sector's Jeans cutoff does **not** alleviate S8 — short by ~35× at the $ $8 scale, and the $ $8 window is already closed at the cutoff (`stage23`); and the *reason* the dark sector is needed is now quantified — at the $ $8 scale today a linear perturbation sits at $y \approx 10^{-3}$, deep MOND, where the kernel would multiply gravity ~29×, so **"dark matter exists at full Ω_dm" is structurally required, not a concession** (`stage24`). **v7 changes — the largest POSITIVE structural change of the programme: $a_0(z)$ is DERIVED from the action.** Until v7 the redshift scaling of a_0 was imposed on the theory as a phenomenological law dressed in DESI's CPL parameters — a dressing that was **never self-consistent** with this paper's own w = -1 exact (§1.2), and is here withdrawn and replaced. The derivation (new §1.3, rows 18–19): promote the constant a_0^2 to $\mathbf{a_0^2(Q) = \kappa^2 c^2 G(-K(Q))/c^2}$ — **the MOND scale is the dark sector's PRESSURE** (p = K is an identity in this sector). Zero new parameters; ONE new relation $\mu^2 \Lambda D^2 = M^4$** ***("the Lagrangian vanishes at the DBI wall"), which* **removes *a free parameter (§3: 5 → 4). Today -K = ρΛ*: the committed coefficient to the digit. Into the past the trace excitation's positive pressure climbs the DBI wall and cancels the vacuum's negative pressure: MOND switches off at recombination** as an output**, with w = -1 still exact — the vacuum never rolls; the *total pressure* evolves. An adversarial external pass (Gemini) proved the density promotion ($a_0^2 \propto \rho Q$) *yields a_0 = const or the wrong sign; that theorem is what forces the pressure route, and is credited in row 18. The full perturbation matrix with the new* YQ mixing is DONE (row 19): the cosmological matrix is untouched *exactly*, the no-ghost theorem **transfers** kernel-independently, c_T = 1 survives. Also in v7: the forest-sector correction — stage 15's claim that the Hooper et al. generic-shape likelihood is public was WRONG (announced, never released; all probes 404); the sector's forest suppression is instead **self-computed** by a lognormal mock (0.20% at the binding bin, nuisance-dominated, `stage16`). Still owed after v7: the CLASS re-run with the derived law, the MUSE/MSA-3D re-exam against the bumpless shape, and a *derivation* of $\beta = 1$ (it is **selected**, not derived — non-claim 2e). **v6 changes — an ERRATUM against v5, filed against my own argument.** v5's non-claim 2d rested one of its three steps on a **category error**: it bounded the *export* of the dust's energy from a galaxy by the khronon's own polytropic sound speed (≈ 690 Gyr). *$\nabla_\mu T^{\mu\nu} = 0$ bounds the **total energy**, not the **flux velocity**; the bound on how fast energy can leave a region is **causality**, and AeST has two massless tensor modes at exactly* c *— by this paper's own row 17. An O(c) export channel therefore exists in the theory as written (1 Mpc in 3.3 Myr).* ***That step is withdrawn.*** *Also withdrawn: the claim that the leak rates "lock" at $\rho\Lambda/\rho$_dust = 2.60* — that is merely today's density ratio. **The conclusion survives on stronger, published, transport-independent grounds:** exporting the energy removes it from the *galaxy*, not the *universe*, so the measured global budget decides — a halo-gated conversion needs a converted fraction $\zeta \approx 0.6$ against the **epoch-marginalised** bound $\zeta < 0.0204$ (Planck) to 0.0374 (+lensing+BAO+SN+DES) [McCarthy & Hill, PRD **108**, 063501], over by 16–29×, and would leave *$\Omega_{dr,0} \approx 0.196 \approx 2100\times$ today's radiation density; the dust→dark-energy channel needs*

far more than the published $|\delta_0| \leq 6.7\times10^{-4}$. And the halo gate is ***adverse****, not protective: dilution goes as a_c, so later injection leaves* more *residue.* ***The lensing theorem (ρ+3p vs ρ+p) is untouched and is what carries the result.*** *Banked as a genuine asset: the Y-gate's FRW exactness makes recombination-era N_eff immunity* exact*. Also recorded, against expectation: the* ***background*** *does not require dark matter at all (Kunz's dark degeneracy, confirmed numerically — a single free w_X(z) reproduces ΛCDM's H(z) with baryons alone); the binding constraint is* ***CMB lensing****, which forces the clustering to survive to $z < 0.30$. Every numbered property below is backed by a committed, runnable script that exits non-zero on failure. Scripts named at each line.* ***v5 changes — the largest revision, and it runs against the programme's interest.*** *A six-stage sequence (`nbody_2026/`, every stage committed and green) settles the galaxy-interior question that v3 reopened as non-claim 2d, and* ***settles it adversely: the Q-sector cannot also be absent from galaxies, and the repair space of local, energy-preserving modifications is provably empty.*** *Three results carry it: (i)* the dust mass IS the conserved shift charge *($\rho = Q_0 n$ with $\rho/n = Q_0$ regardless of any added Y-dependent Q-mass), so it cannot be suppressed locally — which retro-explains three withdrawn mechanisms; (ii)* unlocking the charge does not unlock the energy ($\nabla\mu$T$\hat{}\mu\nu = 0$ holds regardless — though v6 withdraws the transport step that accompanied this, see the v6 banner); (iii) ** dynamics responds to ρ + 3p but lensing responds to ρ + p, and no equation of state kills both** — so the same no-slip identity that underwrites row 1 forbids hiding the dust. Also corrected here: v3/v4's claim that the DBI cap bounds the dust *density* (it bounds the pressure; *ρexc is linear in u and diverges at saturation), and v4's spliced amplitude band.* ***Rows 1–12 and 15–17 are untouched. v2 changes:*** *row 13's shift-charge cluster route is* ***withdrawn*** *(killed by the 1-Mpc confrontation: smooth accretion restores $\$ \to 1 for any cold IC) and replaced by the * *a{0}−bump environment response * *, which passed a five−environment matrix AND its perturbation health check; the health check carved two hard const* $\leq 8.4\times10^{-7}$; amplitude $\leq 2.7\times$ fiducial). The coefficient row now reads as **measured**: $\kappa = 0.551 \pm 0.043$, ± 0.063 if the H_0 tension is carried as a systematic ($\kappa \propto a_0/H_0$). **v3 changes:** (i) the **full AeST perturbation matrix** for the bump term is now DONE and added as row 17 — c_T = 1 exact, no-ghost as a theorem, quasi-static closure derived, amplitude cap softened to a band; non-claim 2c is updated accordingly. (ii) **Non-claim 2 is corrected**: v2's "by item 13 it is absent where rotation curves are measured" was a dangling pointer — row 13 was withdrawn *in the same version* — so the galaxy-interior status of the Q-sector dust is **REOPENED** and now stated as its own open problem (non-claim 2d). v2 carried this internal inconsistency for several hours; the correction runs against the framework's interest and is flagged rather than hidden. (iii) §3's amplitude cap now states its units (4.5 Mpc^{-2} = 2.7× the fiducial 1.65 Mpc^{-2}).

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1. The action

$$S = \int d^4x\,\sqrt{-g}\,\Big[\ \underbrace{\frac{R-2\Lambda_{\rm bare}}{16\pi G}}_{\text{Einstein–Hilbert}} \ +\ \underbrace{\mathcal{L}_{\rm aether}[A^\mu, g]}_{\text{unit-timelike } A^\mu A_\mu=-1} \ +\ \underbrace{\mathcal{F}(Y,Q)}_{\text{the dark sector}}\ \Big] \ +\ S_{\rm matter}[g,\psi]$$

with the two scalar invariants built from the khronon φ and the aether A$\hat{}\mu$,

$$Q \equiv A^\mu\nabla_\mu\varphi, \qquad Y \equiv (g^{\mu\nu}+A^\mu A^\nu)\nabla_\mu\varphi\,\nabla_\nu\varphi.$$

_aether is AeST's FULL aether sector, written out because v7's SVT campaign (row 20) proved that leaving it schematic invites silent truncation:

$$\mathcal{L}_{\rm aether} = \frac{1}{16\pi\tilde{G}}\Big[-\tfrac{K_B}{2}F^{\mu\nu}F_{\mu\nu} + 2(2{-}K_B)\,J^\mu\nabla_\mu\varphi - (2{-}K_B)\,Y - \mathcal{F}(y,Q) - \lambda\,(A^\mu A_\mu{+}1)\Big], \qquad J^\mu = A^\nu\nabla_\nu A^\mu$$

** CORRECTED 2026-08-17.** This block previously had `+ (A·A+1)` (wrong sign), omitted `- ( , )`, and the surrounding text placed *outside* the $1/16\pi$G prefactor and *added* rather than subtracted. Verified against the authors' own LaTeX — arXiv:2007.00082 `\label{NT_A_action}` (Eq. 5) and its independent restatement arXiv:2109.13287 Eq. (1), which agree term by term. The four coefficients this file already had right are -(K_B/2)F^2, +2(2-K_B)J·∇φ, -(2-K_B), and R inside the prefactor. `real_research/bridge1_aest_equations.md` was correct all along and is the reference transcription. See `RETRACTIONS.md`.

The two **(2-K_B) scalar-coupled terms are part of AeST's kernel** (SZ Eq. 5) and are load-bearing — they generate the committed quasi-static G = (1-K_B/2)$\hat{\rm G}$, and omitting them changes the scalar and vector spectra (row 20 caught exactly this in an intermediate derivation and killed the artifacts).

This is **Aether-Scalar-Tensor** (Skordis & Złośnik 2021, PRL **127** 161302) — chosen because it is the only relativistic MOND-class theory that reproduces the CMB. The framework's content is entirely in the choice of the free function , which **factorises**:

$$\boxed{\mathcal{F}(Y,Q) \;=\; \underbrace{\frac{a_0^2}{8\pi G}\,\mathcal{F}_Y\!\left(\frac{Y}{a_0^2}\right)}_{\text{MOND}} \;+\; \underbrace{K(Q)}_{\text{dark energy + dark matter}} \;+\; \underbrace{A\,B\!\left(\tfrac{Y}{a_0^2}\right)(Q-Q_0)^2}_{a_0\text{-bump response (v2)}}}$$

$$B(y) = \frac{y}{(1+y)^2}$$

The third term (v2) is a **position-dependent Helmholtz mass $\mu^2_{\rm eff} = A\,B(g^2/a_0^2)$ peaked at the framework's own a_0** — the location costs nothing (Y/a_0^2 is already the Y-sector's normalisation); one new calibrated amplitude $A \approx 1.7$ Mpc^{-2}. *The dark sector's response is resonant at the MOND transition, and clusters are the cosmic objects that live at a_0* (R_{500} at 0.33–0.58 a_0).

1.1 The Y-sector — MOND

_Y is fixed by the framework's Route A kernel ν(y) = 1/(1 - e^(-$\sqrt{}$y)), y = g_bar/a$_0$. In AQUAL variables it is the closed parametric pair

$$\mu(u) = 1 - e^{-u}, \qquad x(u) = \frac{u^2}{\mu(u)}, \qquad u = \sqrt{y},\; x = g_{\rm obs}/a_0.$$

with the normalisation

$$a_0 \;=\; \kappa\, c\sqrt{G\rho_\Lambda} \;=\; \frac{cH_\Lambda}{Z} \;=\; c^2\sqrt{\frac{\Lambda}{32\pi}} \;=\; 9.3619\times10^{-11}\ \mathrm{m\,s^{-2}}, \qquad \kappa = \tfrac{1}{2},\;\; Z = 2\sqrt{8\pi/3}.$$

$\kappa = \frac{1}{2}$ is FITTED, not derived. See §4.

1.2 The Q-sector — an offset DBI khronon

$$K(Q) \;=\; -M^4 \;+\; \mu^2\Lambda_D^2\left[1-\sqrt{1-\frac{u^2}{\Lambda_D^2}}\right], \qquad u \equiv Q-Q_0, \quad |u|<\Lambda_D$$

Three jobs from one function:

piece	limit	what it is
the **offset** -M^4	u = 0, K = 0	ρ = -K = M^4, p = K = -M^4 $\Longrightarrow$ **w = -1 exactly**: dark energy
the **deviations**	u $\ll$ Λ_D	K $\to \mu^2$u^2/2 is the **pressure**, while ρ_exc = Q$_0\mu^2$u is **linear** in u $\Longrightarrow$ w = u/2Q$_0$ $\to$ 0: **dust**, i.e. the dark matter. *(v5: the linearity is why the dust mass equals the conserved charge — see non-claim 2d.)*
the **saturation**	u $\to$ Λ_D	K bounded, K $\to \infty \Longrightarrow$ w $\to$ 0 again: kills the early stiff phase

Parameters: **M** (vacuum scale, M^4 = ρ_Λ), μ (Helmholtz mass), **Λ_D** (DBI field-space scale), **I**$_0$ (the conserved shift charge = the dust amount). **(v7: Λ_D is no longer free — μ^2Λ_D^2 = M^4, §1.3.)**

1.3 a$_0$(z) — derived, not imposed (v7)

The constant a$_0^2$ in §1.1 is promoted to a scalar of the Q-sector:

$$a_0^2(Q) \;=\; \frac{\kappa^2 c^2\, G\,(-K(Q))}{c^2}\,, \qquad \text{i.e.} \qquad a_0 = \kappa c\sqrt{\frac{G\,(-p_Q)}{c^2}}$$

The MOND scale is the dark sector's pressure — and since p = K identically (§1.2), this is the unique algebraic scalar of Q that (i) equals ρ_Λ today (u $\to$ 0: -K = M^4, reproducing §1.1's normalisation to the digit) and (ii) *declines* into the past. The competing candidate, a$_0^2 \propto \rho$_Q, is **excluded by one bit**: the density grows with excitation, so a$_0$ would rise toward recombination — MOND ON at the CMB (the adversarial theorem credited in row 18 proved that route also gives a$_0$ = const on the pure-vacuum branch; either way it is dead). The pressure route additionally has charge-suppressed backreaction: da$_0^2$/dQ $\propto$ -K = -n, versus the density route's QK, unsuppressed.

With **$\beta\,\mu^2$Λ_D^2/M^4 = 1** — the Lagrangian vanishes exactly at the DBI wall — the background law is closed-form (n $\propto$ a^{-3} exactly; ν n/(μ^2Λ_D)):

$$\frac{a_0^2(z)}{a_0^2(0)} = \frac{\sqrt{1+\nu_0^2}}{\sqrt{1+\nu_0^2(1+z)^6}}$$

constant to <1% through every MOND test (z ≤ 5), transition at z_t = ν_0^(-1/3) - 1 ∈ [17, 35], off at recombination (0.002–0.006). **w = -1 stays exact throughout** — the vacuum never rolls; the *trace* excitation ($\Omega \leq 4.4 \times 10^{-7}$, the stage-3 black-hole ceiling) climbs the wall and its pressure reaches M^4 regardless of its amount, because the wall height is a property of the potential, not the charge. The window $\nu_0 \in [2.1 \times 10^{-5},\ 1.8 \times 10^{-4}]$ is cut from *below* by the CMB off-switch and from *above* by the RAR — and the RAR ceiling exists only because the captured charge **drains** (stages 2–3): had it formed a halo, the local a_0 would fall to 0.55 of nominal at the $g = a_0$ radius, RAR-fatal. **The adverse drain result of non-claim 2d is load-bearing for this derivation.** The old CPL-dressed law (bump at z ≈ 0.5) is withdrawn; the derived law has no bump. Scripts: `nbody_2026/stage17_a0z_from_the_action_2026.py` (17/17), `nbody_2026/stage18_perturbation_matrix_with_mixing_2026.py` (10/10).

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2. What is verified

#	property	value / statement	script
1	**Lensing** — the test that killed modified inertia	$\Phi = \Psi \Longrightarrow \gamma$_PPN = 1, M_dyn/M_lens = 1 exactly. **21.2σ → 0.601σ**	`mi_relativistic_completion_aest_`
2	**Kernel embeds** in _Y	deep-MOND $\mu \to$x exact; Newtonian $ →1; $x(u)abijection(h = e^(-u)(1+u) > 0); freefunction ∗ ∗convex ∗ ∗\|same\|\|3\| ∗ ∗Newtonianresidual ∗ ∗\|e^(−√$y) = 3.6×10^{-3457} at Earth's orbit	same
4	**Dark energy is not an input**	w = -1 **exactly** at the condensate minimum — proved, not assumed	`mi_condensate_vacuum_energy_a0_2`
5	**Ghost-free** over the whole field range	K = μ^2(1-s^2)^(-3/2) > 0 for all |s|<1	`mi_dbi_khronon_2026.py`
6	**Subluminal** everywhere	c_s^2 = Λ_D s(1-s^2)/(1+Λ_D s) ≤ 0.385 Λ_D	same

#	property	value / statement	script
7	**The published no-go is dissolved**	Blanchet & Skordis 2024 §4.3.1's 455× conflict **reverses sign**: bounded pressure $\Longrightarrow$ w$→$0 early $\Longrightarrow$ cosmology becomes a *lower* bound on μ^{-1}, same direction as MOND	same
8	**Theorem**: why no polynomial works	for K ~ u^n, early-time w $\to$ 1/(n-1). Quadratic 1, quartic $\frac{1}{3}$, sextic — all fail. **Only boundedness gives w$→$0**	same
9	**CMB acoustic peaks** — real CLASS run	at Λ_D $\leq 10^{-2}$, c_s^2(recomb) = 2.9×10^{-8}; TT unchanged to **0.069%**, P(k=0.2) to 1.71%. Indistinguishable from CDM	`mi_dbi_cmb_class_run_2026.py`
10	**BTFR**	exact, from convexity	`mi_route_a_field_theory_2026.py`
11	**g^{-2} Lorentz violation**	**restored** by the aether (pure Bekenstein–Milgrom had lost it)	`mi_relativistic_completion_aest`
12	**a$_0$ tied to Λ**	a$_0$ = m_cond/(4$\sqrt{\pi}$) with m_cond = M^2/($\sqrt{2}M_P l$), $M^{4} = ρ_Λ — agrees to **0.076%**	`mi_condensate_vacuum_energy_a0_2`

#	property	value / statement	script
13	**Clusters** (v2)	~~shift-charge IC route~~ **withdrawn**: killed by the 1-Mpc confrontation — cluster R500 and galaxy-outskirt lensing share a scale with disjoint requirements, correlated Gaussian ICs cannot select environments, and **smooth accretion restores** **$ $\rightarrow 1 for any cold IC$ * *. Replaced by the $ * *a{0}-bump response$ * $(row 15)\|`mi_c_route_1 mpc_ untouched by_2026.py`(9/9)\|\|14\|$ * $ * Directional EFE signal$ * $ * |pure MI predicted$ * $exactly zero$ *; $AQUAL-class predicts 1–4\|15\|$ * $ * a{0}$-bump environment response**	passes all five environments: cluster (calibration, μ^2_eff = 0.23 Mpc^{-2}), galaxy interior 0.034% of M_b (RAR cost 4×10^{-4} dex), galaxy 1 Mpc 1.2% vs the 5.9% strict bound, linear cosmos 0.6% of mean matter, solar 10^{-19}. Vanishes on FRW (Y=0) and is second-order in perturbations ⇒ linear CMB/P(k) unaffected by construction. Evades both prior kills (a response has no charge to advect; couples to environment directly)
16	**...and its health check**	**no ghosts ever** (kinetic contribution 2AB ≥ 0, = 0 on FRW), no Ostrogradsky sector; FRW gradient health **excludes the old Λ_D = 10^{-2} and forces Λ_D $\leq$ 8.4×10^{-7}** (window stays 3.7 orders); halo gradient health **caps A $\leq$ 2.7× fiducial, excluding the demanding end of Mistele's cluster band** — a two-sided falsifiable pinch; free signature: cluster anisotropic stress at O(0.6)	`mi_a0_bump_health_2026.py` (11/11)

#	property	value / statement	script
17	**...and the FULL perturbation matrix (v3)**	with all aether mixings: **c_T = 1 exact** (the new terms are algebraic in the metric — GW170817 safe by construction); **no-ghost is a THEOREM** (the bump's kinetic addition is rank-1 PSD in the $\chi\chi$ entry $\Longrightarrow$ Weyl monotonicity: it cannot lower any eigenvalue of AeST's healthy kinetic matrix; closed-form 2× 2 + 200 random 4× 4); unit-norm gives $\delta A^0 = -\Phi A^0 \Longrightarrow \delta Q = \chi - Q_0\Phi$ — the quasi-static phenomenology of row 15 is **derived, not assumed**; gradient block exact: $\Delta G = 2\lambda$[[q, 2y^{3/2}B],[2y^{3/2}B, yq]], q = $(3y^2-8y+1)/(1+y)^4$; integrating out the aether softens the amplitude cap: at base_a = 1 the band is **A_max ∈ [2.72, 4.46]× fiducial = [4.49, 7.36] Mpc^{-2}** (**v4 correction:** the earlier "(2.7–7.4)× fiducial" spliced fiducial units with Mpc^{-2}, making the band look 1.66× wider at the top than it is — base_a_lookup_and_robustness_2026.py, 12/12). **The base_a lookup is now DONE and it converts the caveat rather than closing it:** base_a is built	mi_aest_full_matrix_bump_2026.py (11/11)

#	property	value / statement	script
18	**$a_0(z)$ DERIVED from the action (v7)**	$a_0^2(Q) = \kappa^2 c^2 G(-K(Q))/c^2$ — the MOND scale is the dark sector's **pressure**. Zero new parameters; one relation $\beta = \mu^2 \Lambda_D^2/M^4 = 1$ (**selected** by the CMB off-switch, not derived — non-claim 2e). Today's coefficient reproduced to the digit; **w = -1 stays exact** (the vacuum never rolls; the total pressure evolves). Density promotion **excluded by one bit** (a_0 would rise into the past); the exclusion theorem for the naive route is due to an adversarial external pass (Gemini, 2026-08-10) and is credited as such. Closed-form law: $a_0^2(z)/a_0^2(0) = \sqrt{(1+\nu_0^2)}/\sqrt{(1+\nu_0^2(1+z)^6)}$; constant to <1% for $z \leq 5$; transition $z_t \in [17, 35]$; off at recombination (0.002–0.006). Window $\nu_0 \in [2.1 \times 10^{-5}, 1.8 \times 10^{-4}]$, cut by the CMB from below and the RAR-via-drain from above — **non-claim 2d's adverse result is load-bearing here**. The CPL-dressed law (never self-consistent with w = -1 exact) is withdrawn	`stage17_a0z_from_the_action_2026` (17/17)

#	property	value / statement	script
19	**...and ITS full perturbation matrix (v7)**	the promotion's _YQ $\neq 0$ mixing analysed: **the cosmological matrix is untouched EXACTLY** ($h^{00} = g^{00} + A^0A^0 = 0$ exactly **for A^i = 0** — with the aether scalar on, $h^{00} = (\partial\alpha)^2/a^2$ at *second* order (row 20 refinement); $\delta Y^1 = 0$ identically either way; every new entry carries a positive power of y, and $y = 0$ on FRW — the CLASS pass, k^4-Jeans structure and GDM degeneracy theorem inherit without recomputation); **the no-ghost theorem TRANSFERS kernel-independently** (both kinetic additions land in the same $\chi\chi$ slot as row 17's: (F-yF) ≥ 0 because ghost-freedom of the base meets concavity of the kernel, and (^{2}/)y^2F is a perfect square with $\propto$ -K = $M^4/\sqrt{(1+\nu^2)}$ > 0 always — a_0^2 can never cross zero; Weyl monotonicity does the rest); the mixing never reaches the kinetic matrix (δY has no time derivative), its drift dispersion is real at every k, and its gradient/force corrections are trace-bounded at $1.3 \times 10^{-4}/1.7 \times 10^{-4}$ of committed entries; c_T = 1 survives	`stage18_perturbation_matrix_with` (10/10)
11			

#	property	value / statement	script
20	**...and the COVARIANT SVT DECOMPOSITION (v7, post-publication)**	nine-agent derive-and-adversarially-verify campaign; 17 executed sympy scripts committed in `nbody_2026/svt_2026/`, all exit 0. **CONFIRMED under the full action**: c_T = 1 exact from explicit $O(h^2)$ components $((a^3/32\pi G)^2$ - $(a/32\pi G)(\partial h)^2$ per polarisation; gravitons massless on the background shell; independent Euler route agrees); **the promotion drops out of FRW perturbations at second order — the whole promoted MOND term starts at THIRD order** (verified with generic (Q) and live chain rule); $\delta A^0 = -\Phi + (3/2)\Phi^2 + (\partial\alpha)^2/2a^2$ exact; $\delta Y^2 = \lvert\partial(\chi+Q\alpha)\rvert^2/a^2$, a perfect square in the shift-invariant combination; the quasi-static $\chi\chi$ block exact (unchanged by $\sqrt{}$(-g) and second-order $\delta Q/\delta Y$ pieces); drift asymmetry closed form $D^2/(G_KG_G) = (\kappa^2/4\pi)s^2y^3F^2/(F+2yF)$, ceiling 5.1×10^{-3}; corrected coefficients: second kinetic addition $\leq 0.0267\kappa^2 s^2 K$, q-window (0.1315, 2.5352). **THE**	`stage22_covariant_svt_decomposit` (20/20, drives the full suite)

3. Parameter count, honestly

	ΛCDM	this completion
dark sector	Ωc, $\Omega\Lambda$	M (= ρ_Λ^{1/4}), I_0, μ, Λ_D **(v7: Λ_D = M^2/μ, no longer free), A (v2)**
MOND scale	—	a_0 — **not independent**, = m_cond/(4$\sqrt{\pi}$) via item 12; **(v7) and its z-dependence is now derived too (§1.3)**
bounds	—	Λ_D: $1.9 \times 10^{-10} \ll$ Λ_D $\leq 8.4 \times 10^{-7}$ (**health-bounded, v2**; 3.7 orders) — **(v7) now fixed at M^2/μ by $\beta = 1$, which must land inside this window: a consistency check inherited rather than retired**; A ≈ 1.7 Mpc^{-2} calibrated on clusters, **health cap A $\leq$ 4.5 Mpc^{-2} = 2.7× fiducial** (row 16; the full matrix widens it to [2.72, 4.46]× fiducial = [4.49, 7.36] Mpc^{-2} at base_a = 1 — row 17, where the base_a fork and its robustness are stated); μ^{-1} = 4392 Mpc (item 12); $I_0 \approx$ Ω_dm (an IC); **(v7) the khronon's own trace charge $\nu_0 \in [2.1 \times 10^{-5}, 1.8 \times 10^{-4}]$ (§1.3)**

So: **(v7) four dark-sector numbers against ΛCDM's two** (v2 added A; v7 removes Λ_D via $\beta = 1$), with a_0 — value *and* redshift dependence — derived from them rather than added. Still not fewer parameters than ΛCDM. What it buys is that Λ, dark matter, MOND, *and now the evolution of the MOND scale* come from **one function** instead of unrelated sectors.

4. What is NOT claimed — read this before quoting anything above

1. $\kappa = \frac{1}{2}$ **IS NOT DERIVED.** Item 12 looks like a derivation and is not. `a = m_cond/(4√)` is *algebraically identical* to `a = ½√(G _Λ)` and to `Z = 2√(8/3)` — three costumes, one statement — and the residue is **convention-dependent** (4$\sqrt{\pi}$ with the reduced Planck mass,

$√$2 with the non-reduced). A derived number cannot depend on a bookkeeping choice. It is a **relabelling**, as this corpus's own theorem already established. What genuinely moved: "why $\kappa = \frac{1}{2}$?" becomes "why is $a_0 = m_cond/(4\sqrt{\pi})$?", which is sharper because m_cond is a *derived* quantity of a real field theory rather than a definition.

2. **Dark matter EXISTS**, at the full Ω_dm. Removing the pressureless component moves H_3/H_1 by 54% and no refit absorbs it ($\Delta\chi^2 > 400$). What does **not** exist is a dark-matter *particle*: it is the Q-sector of the same scalar that supplies Λ and MOND. **(v3 correction:** v2 continued "and by item 13 it is absent where rotation curves are measured" — a dangling pointer, since v2 itself withdrew row 13. See 2d. **The defensible slogan shrinks to "no dark-matter particle" — never "no dark matter"; and (v5)** "none in galaxies" is WITHDRAWN as a claim of this theory**, not merely open — see 2d.)** 2d. **(v5) THE Q-SECTOR CANNOT ALSO BE ABSENT FROM GALAXIES — FALSIFIED WITHIN THE THEORY AS WRITTEN, AND THE REPAIR SPACE IS NOW PROVABLY EMPTY.** This is the largest change since v1 and it runs against the programme's interest. v2 asserted "none in galaxies" via a pointer to withdrawn row 13; a six-stage sequence (`nbody_2026/`, all scripts committed, 20/20 + 15/15 + 12/12 + 15/15
   - 15/15 + 11/11) then settled the question the pointer had concealed.

   The physics. The 1-Mpc smooth-accretion theorem implies the Q-sector dust falls into *every* collapsing basin, galaxies included. What happens next was the open problem, and it closed in four steps:
   - **Nothing stops the collapse.** Wave (k^4 ghost-condensate) pressure gives a soliton scale of **0.18 AU** at halo density — eight orders below the RAR region — and it *shrinks* as ρ grows ($\lambda \propto \rho^{\widehat{}}\text{-}\frac{1}{4}$); a 1 kpc wave core would need M below the Lyman-α fuzzy-DM floor. The dust *is* an n = 1 polytrope ($p = K\rho^2$, $c_s^2 = 2K\rho$ rising) with a genuine **mass-independent 105 pc** equilibrium — but that lies **14.4× past DBI saturation**, where c_s → 0 and the stiffening switches off. Endpoint: a black hole of the captured share, **falsified** $5.8 \times 10^5\times$ **against Sgr A*** (4.3×10^6 M from individual stellar orbits — no mass model, kernel or cosmology in the comparison). *(v5 corrects v3/v4 here: an earlier draft claimed the DBI cap bounds the density and yields a ~250 pc core. $\mu^2u^2/2$ is the **pressure**; exactly, $\rho_exc = Q_0\mu^2 u$ is **linear** in u and diverges at saturation. Withdrawn.)*
   - **THEOREM: the dust mass IS the conserved shift charge.** $\rho = Q_0 \cdot n$ with n the shift-charge density, and **$\rho/n = Q_0$ independently of any Y-dependent Q-mass, of any shape or amplitude.** So no such term can change how much dust a galaxy carries — *a conserved charge cannot be suppressed locally, only moved (gravity moves it inward) or not conserved.* This single statement retro-explains three withdrawn mechanisms: the shift-charge IC route, the Φ^n response, and an acceleration-gated suppressor built and killed in stage 4–5 (its eight gates all tested the field amplitude u; the observable is n, and at fixed charge the suppressor makes a region a *cheaper* place to store charge — it attracts what it was built to expel).
   - **Breaking the shift symmetry gets furthest, and fails.** A *global* potential is incompatible as an identity ($dV/dt = V\varphi$ must vanish for w = -1, while the condensate needs $\varphi = Q_0 \neq 0$); quantitatively the leak rates lock at $\Gamma n/\Gamma V = \rho\Lambda/\rho\text{dust} = 2.60$, so clearing a galaxy in 1 Gyr drifts $\rho\Lambda$ *by 47× and protecting* $\rho\Lambda$ forces a 180 Gyr charge lifetime. A **Y-gated** breaking, $-W(Y/a_0^2)V(\varphi)$ with $W(0) = 0$, *defeats* that — on FRW the symmetry is **exact**, so cosmology is untouched identically. But **unlocking the charge does not unlock the energy**: $\nabla_\mu T^{\widehat{}}\mu\nu = 0$ holds regardless. **(v6 erratum:**

v5 added that the energy therefore *cannot leave*, bounding its export by the khronon's polytropic sound speed at ≈ 690 Gyr. That was a category error — a conservation law bounds total energy, not flux velocity; the causal limit is c, and row 17's tensor modes travel at exactly c. **Withdrawn.** What kills the export route instead is the *measured global budget*: a halo-gated conversion needs $\zeta \approx 0.6$ against the epoch-marginalised $\zeta < 0.0204$–0.0374 [McCarthy & Hill, PRD **108**, 063501] — over by 16–29× — leaving $\Omega_dr,0 \approx 0.196$, some 2100× today's radiation density; and dust→ Λ needs far more than $|\delta_0| \leq 6.7 \times 10^{-4}$. The gate is adverse rather than protective, since dilution goes as a_c.**)

- ** AND LENSING CLOSES IT BY A THEOREM.** Transforming the equation of state almost works: the dynamical source $\rho + 3p$ vanishes at exactly $\mathbf{f = 1/3}$, and the field-strength gate makes that a *stable fixed point* rather than a tuning. But **dynamics responds to $\rho + 3p$ while lensing responds to $\rho + p$** — at f = 1/3 the lensing source is still $(2/3)\rho$, giving M_lens/M_dyn ≈ 29 against an observed 1.0–1.3. Generally, $\rho + 3p = 0$ requires w = -1/3 and $\rho + p = 0$ requires w = -1: **incompatible. No equation of state renders a given energy density invisible to both orbits and light; the only such configuration is $\rho = 0$**, which the transport bound forbids reaching. *The no-slip identity $\Phi = \Psi$ that makes this theory's modified-gravity arm work — the property that killed its modified-inertia arm at 21σ and underwrites item 1 — is the same property that forbids hiding the dust.*

Status, stated without spin. The galaxy-interior problem is not a missing mechanism: it is a conservation law (energy, via general covariance), plus a transport bound, plus lensing. **The repair space of local, energy-preserving modifications is provably empty.** The defensible slogan is therefore **"no dark-matter particle"** and nothing more; **"none in galaxies" is withdrawn as a claim of this theory.** What remains open is a structural change — the Q-sector not acquiring the galactic energy in the first place — against which the smooth-accretion theorem stands for any cold initial condition. Items 1–12 and 15–17 are untouched by all of this.

2e. **(v7) THE $a_0(z)$ DERIVATION'S OWN CAVEATS, stated before anyone quotes row 18 as finished.** (i) **$\beta = 1$ is BOUNDARY-PINNED** *(upgraded 2026-08-10 from "selected" by `nbody_2026/stage20_beta_equals_one_derivation_2026.py`, 8/8).* Three results: **$\beta \leq 1$ is a THEOREM** — an additive constant in K is Λ_bare in disguise (a redundancy the promotion breaks), and any $\beta > 1$ makes a_0^2 cross zero at finite redshift, where the kernel goes complex for any spatial khronon gradient ($\beta = 1.01$ dies before recombination; β-1 = 10^{-8} by z ~ 10^4); **$\beta = 1$ K is a PURE brane action** (the offset-DBI decomposes exactly into tension × volume element + constant; the constant-free form's zero at the wall is canonical — a null brane has zero action); and the CMB pins **1-$\beta \leq 3.6 \times 10^{-5}$** from below. The defensible sentence: *"$\beta \leq 1$ is derived; the boundary is canonical under the brane reading; the CMB pins the theory to within 3.6×10^{-5} of it."* Anyone quoting "$\beta = 1$ is derived" is overquoting. Named escapes, kept open: $\beta = 1$-ε is consistent with a falsifiable high-z MOND floor $\sqrt{\varepsilon \cdot a_0}$ (a detection would *measure* 1-β); and the $\beta \leq 1$ theorem needs the charge history to extend to arbitrarily early times. (ii) **The DESI tension is not resolved and cannot be resolved by this sector**: the non-dust remainder's w(z) ≥ -1 always, so DESI's preferred phantom past ($w_0 + w_a = -1.61$) is unreachable. If DESI's evolving-w detection stands, it remains evidence against w = -1 exact. What v7 fixes is only that the $a_0(z)$ law no longer *depends* on borrowing DESI's numbers. (iii) ~~The CLASS re-run with the derived law~~ — **CLOSED 2026-08-10 by `nbody_2026/stage19_class_rerun_derived_law_2026.py`**

(8/8): the exact $\beta = 1$ background is $\rho nd = M^4\sqrt{(1+\nu^2)}$, $p_nd = -M^4/\sqrt{(1+\nu^2)}$ ($\rho p = -M^8$ *an exact invariant) — vs ΛCDM one cold* a^{-3} *trace (today-density* $\rho\Lambda\nu_0 \leq 1.2 \times 10^{-4}\rho_$crit) and an a^{-6} transient $< 3.2 \times 10^{-8}$; θ^* moves 0.18× Planck precision; real CLASS runs give $\Delta\chi^2 = 1.3$ against cosmic variance over 4998 TT+EE multipoles (0.01σ — an ideal experiment could not distinguish them); the MOND-off input of the committed pass is now an output, $a_0(z^*) = 0.002$–0.006 of today. Same rigor level as the committed pass (stock CLASS, trace bracketed as cold matter — exact at high z, conservative at low z). ~~The MUSE/MSA-3D re-examination~~ — **CLOSED 2026-08-10 by `nbody_2026/stage21_muse_msa_reexam_derived_law_2026.py`** **(7/7)**: the MUSE tension *persists* (15.1σ raw, ~2–3σ after the shared ΛCDM assembly drift) but re-grades from framework-distinctive-WORST (the withdrawn declining branch: wrong sign, 16.3σ) to **shared-with-all-constant-a_0-MOND**; in the committed joint fork likelihood the framework's branch relabels M-DEC → M-FLAT (the branch needing exactly the drift MSA-3D measures, p = 1.22, and its near-unfalsifiability critique inverts into the sharp null below); MSA-3D re-grades WEAK-TENSION/WATCH → CONSISTENT (1.15σ from flat). **Forfeited, priced**: the z ≈ 0.4 bump defense; the 3.1–4.2σ DESI cosmology tailwind (and with it the w → -1 hostage — dissolution is now the *prediction*); the z ≈ 2–3 declining discriminator. **New exposure**: full-strength MOND through cosmic dawn ($a_0(10) = 0.99$–1.00 vs the old 0.36) — Nusser-type over-production unpriced, $z_t \in [17, 35]$ in the 21-cm window; the early-structure front needs its own confrontation. ~~The covariant SVT decomposition~~ — **CLOSED 2026-08-10 by row 20** (`nbody_2026/stage22_covariant_svt_decomposition_2026.py`, 20/20, driving 17 committed executed scripts): every v7-specific claim survived adversarial verification under the full action; the truncated-action artifacts an intermediate derivation produced were caught and killed by the verifiers before entering the corpus. **Still owed** (base-AeST, not v7): the in-repo full-action FRW scalar spectrum + the k^4 soft-mode WATCH confronted with AeST's published stability discussion; and the cosmic-dawn confrontation opened by the stage-21 re-exam. *(The β = 1 derivation moved to (i): boundary-pinned, 2026-08-10.)*

3. **The post-recombination growth history is NOT verified.** Item 9 covers the acoustic peaks only. Holding c_s^2 at its peak value for all time *destroys* P(k=0.2), so the pass depends essentially on the bump being transient (it peaks at z ≈ 189). **A patched CLASS fluid carrying $c_s^2(a)$ is required, not optional.**
4. **(v4 rewording — the original referred to withdrawn row 13.)** The dust component's small-scale initial conditions are **not confronted with Lyman-α.** The withdrawn IC route needed a khronon transfer function T ≈ 0.33 at k ≈ 4.5 Mpc^{-1} (ΛCDM-excluded); the surviving picture (standard cold dust + the bump response) should be Ly-α-safe by construction, but that is **asserted, not computed — the confrontation remains owed item in the programme.**
5. **Cassini Q_2 is inherited** (3–15σ) and is not relieved by anything here.
6. **PPN preferred-frame parameters** α_1, α_2 for this have not been computed against lunar-laser and binary-pulsar bounds.
7. **The wide-binary target is the point-field asymptote only.** $\gamma_v = \sqrt{\nu}(y_extN)$; the full nonlinear AQUAL-EFE solve is owed.
8. **This is not a theory of everything** and no part of it addresses the Standard Model. The 2026-06-23 retraction stands.

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5. The falsifiable predictions

prediction	value	how to test
wide-binary boost	γ_v = $\sqrt{\nu}$(y_extN)	**Gaia DR4**, pre-registered and hash-stamped
directional EFE asymmetry	1–4%, signed	already firing at p = 0.029 with the right sign
cluster-to-cluster **scatter** in the residual	$\sigma(\xi)/\xi \approx 0.02$–$0.60$	existing cluster samples — **(v4) separates the a_0-bump response (scatter tracks each cluster's distance from the resonance, rows 15–17) from any fixed-scale mechanism, which predicts uniformity** (the two prior candidates are withdrawn)
cluster mass **profile** misfit	ρ_c flat, M_c ~ r^3, can go negative	fit clusters with the khronon profile instead of NFW
g^{-2} Lorentz violation	computable s_$\mu\nu$	SME sector bounds

prediction	value	how to test
accelerated structure formation — RESTORED (v9)	**collapse is 1.34–1.96× faster than Newtonian, from the a_0-line alone, zero free parameters ⇒** objects ΛCDM forms at z = 10 appear at **z ≈ 14–16**. *(v9 erratum: an earlier v9 draft WITHDREW this row on two wrong grounds, both mine. (a) "It is a linear-growth claim, and rows 19–20 make linear growth ΛCDM's" — a misattribution: collapse TIMING is **nonlinear**, the a_0-line governs accelerations at every scale, and the $\delta Y^1 = 0$ theorem is silent about nonlinear collapse. Both statements hold at once: **ΛCDM halo abundance, accelerated collapse timing.** (b) "The nonlinear boost is only $\lesssim 1.5\times$" — that was the boost at $\delta \approx 200$, the **end** of collapse. At **turnaround**, which is what sets the timing, it is **2.3–5.1**×, i.e. 1.6–1.8× larger in every case. Evaluating a rate-limiting quantity at its minimum is the manufactured-deficit failure this paper's own working rule forbids.)*	resolved high-z galaxy assembly. **And a_0(z) gives it a shape no constant-a_0 MOND has**: the speedup *declines* with redshift — 2.03× at z = 6, 1.66× at z = 10, 1.41× at z = 15, 1.14× at z = 25 — so accelerated formation **switches off** toward the highest z. JWST high-z massive/mature galaxies (McGaugh et al. 2024) test the amplitude; the *decline* is the framework-distinctive part. **Live uncertainty, named:** the boost acts on the **total** enclosed mass, dark sector included, which is this framework's known double-counting exposure (non-claim 2d) — so 1.34–1.96× is an **upper** bound, and the honest range runs down toward ~1 if the dust dominates the collapse (`stage26_collapse_acceleration_own_term` 4/4)

prediction	value	how to test
early-rotator BTFR zero point (v9) — *the prediction that replaces it*	**v(z)/v(0) = [a_0(z)/a_0(0)]^{1/4} at fixed M_b, zero parameters.** $v^4 = G\,M_b\,a_0$ is a THEOREM of §1.1's own a_0-line (its low-g_bar limit is g_obs = $\sqrt{(a_0 g_{bar})}$ exactly, r drops out, d ln v/d ln a_0 = 1/4 exactly). With §1.3's derived a_0(z): rotators lie on **today's** BTFR to **0.009% out to z = 5**, then fall below it by **0.34% (z = 10), 2.6% (z = 15), 7.8% (z = 20), 18.9% (z = 30)** — signed, high-z rotators are *slower* at fixed baryonic mass. Footing-independent (the z = 0 anchor cancels in the ratio)	resolved high-z rotation curves with baryonic masses. **Two-sided:** below z = 5 the prediction is *exactly* today's BTFR, so any robust zero-point offset falsifies the derived law; the deficit first clears the committed 0.06 dex high-z systematic floor (= 3.5% in velocity, since $v^4 \sim M$) at **z ≈ 17**, where it becomes a curve to trace. ALMA [CII]/JWST reach z ~ 4–7 today, so this is a **pre-registered forecast for the 2030s, not present-day support** (`stage25_cosmic_dawn_own_terms_2026.py`, 8/8)
MOND-regime threshold and its evolution (v9)	**$\Sigma = a_0/(\pi G) = 214$ M/pc^2**, derived from the a_0-line's own transition condition g_bar = a_0 — *twice* the $a_0/(2\pi G)$ usually quoted, which belongs to a different interpolation and must not be applied to this framework. And $\Sigma(z) \propto a_0(z)$, so **the MOND regime SHRINKS toward high z: early systems are MORE Newtonian than today's at equal surface density**	any high-z system with measured surface density and kinematics. Note the consequence for the JWST objects that motivated the withdrawn row: compact high-z galaxies (~10^3 M/pc^2) sit ~5× *above* Σ, i.e. in the Newtonian regime where a_0's value does not enter — **they cannot test a_0(z) in either direction**
a_0(z) shape (v7)	**constant to <1% for z ≤ 5, transition z_t ∈ [17, 35], NO bump at z ≈ 0.5**	any measurement of a_0 evolving below z ≈ 5 falsifies the derived law (the withdrawn CPL dressing predicted +6% at z = 0.5 — the two are now distinguishable); MUSE-type high-z kinematics, high-z RAR

6. Credit

$\nu = \sqrt{(1+1/y)}$ is **Milgrom 1999 PLA 253:273 eq 9**. MOND: Milgrom 1983 ApJ 270:365. AQUAL: Bekenstein & Milgrom 1984 ApJ 286:7. TeVeS: Bekenstein 2004 PRD 70:083509. **AeST: Skordis & Złośnik 2021 PRL 127:161302** — the completion itself. Ghost condensate: Arkani-Hamed, Cheng, Luty & Mukohyama 2004 JHEP 0405:074; nonlinear: +Wiseman 2007 JHEP 0701:036. $K(Q) = \mu^2(Q\text{-}1)^2$ and the ghost- condensate identification: Verwayen, Skordis & Złośnik 2024; **Blanchet & Skordis 2024 JCAP 11(2024)040** — including §4.3.1, the no-go this work dissolves. Quasi-static AeST: Durakovic & Skordis 2024 JCAP 04:040; Verwayen, Skordis & Boehm 2024 MNRAS 531:272. **Mistele, McGaugh & Hossenfelder 2023 A&A 676:A100** — the weak-lensing bounds that killed this program's first two cluster mechanisms and now two-sidedly bound the third (rows 15–17). GDM bound: Kopp, Skordis, Thomas & Ilić 2018 PRL 120:221102. Ly-α floor: Rogers & Peiris 2021 PRL 126:071302. RAR scatter: Desmond 2023. Accelerated structure formation: McGaugh, Schombert, Lelli & Franck 2024.

The distinctive content of this work is the coefficient — $a_0 = \kappa c \sqrt{(G\rho_\Lambda)}$ with $\kappa = \frac{1}{2}$ — its embedding in _Y, the offset-DBI Q-sector, and items 7, 8, 9, 12 and 13. Everything else is cited above and is not mine.